



Lower Secondary Computing Part One Programme of Study

Module One: Computational Thinking with Python and Turtle	Study Sessions	Coursebook Pages	National Curriculum Link
Topic One (1.1): Decomposition	Session 1 (1.1.1) – Introduction to Decomposition	Pages 4-5	Subject content covered in this module: • design, use and evaluate computational abstractions that model the state and behaviour of real-world problems and physical systems • use two or more programming languages, at least one of which is textual, to solve a variety of computational problems; make appropriate use of data structures [for example, lists, tables or arrays]; design and develop modular programs that use procedures or functions
	Session 2 (1.1.2) – Applying Decomposition	Pages 5-10	
Topic Two (1.2): Abstraction	Session 1 (1.2.1) – Introduction to Abstraction	Pages 10-13	
	Session 2 (1.2.2) – Applying Abstraction	Page 13-14	
Topic Three (1.3): Pattern Recognition	Session 1 (1.3.1) – Introduction to Pattern Recognition	Pages 15-16	
	Session 2 (1.3.2) – Applying Pattern Recognition	Page 17-19	
Topic Four (1.4): Pattern Recognition with Subroutines	Session 1 (1.4.1) – About Subroutines	Pages 19-20	
	Session 2 (1.4.2) – Subroutines with Parameters	Pages 20-22	
Topic Five (1.5): Algorithms	Session 1 (1.5.1) – About Algorithms	Pages 23-26	
	Session 2 (1.5.2) – Algorithms and Flow Charts	Page 26	





Topic Six (1.6): Challenges and Final Assessment	Session 1 (1.6.1) – Shape Challenges	Pages 27-28	
	Session 2 (1.6.2) – Maze Challenges	Pages 27- 28	
ASSIGNMENT ONE	Computational Thinking	Recall of key terminology, skills in flowcharts, pseudocode and python turtle.	



Module Two: Word Processing	Study Sessions	Coursebook Pages	National Curriculum Link
Topic One (2.1): Introduction to Word Processing	Session 1 (2.1.1) – Writing your First Words	Pages 3-12	Subject content covered in this module: - undertake creative projects that involve selecting, using, and combining multiple applications, preferably across a range of devices, to achieve challenging goals, including collecting and analysing data and meeting the needs of known users - create, re-use, revise and re-purpose digital artefacts for a given audience, with attention to trustworthiness, design and usability
	Session 2 (2.1.2) – Basic formatting and fonts	Pages 12-16	
Topic Two (2.2): Text Formatting and Alignment	Session 1 (2.2.1) – Paragraphs, Alignment and Lists	Pages 16-21	
	Session 2 (2.2.2) – Indentation, Spacing, Borders and Shading	Pages 21-26	
Topic Three (2.3): Using Graphics and Tables	Session 1 (2.3.1) – Images	Pages 26-29	
	Session 2 (2.3.2) – Tables and Smart Art	Pages 29-34	
Topic Four (2.4): Advanced Formatting Techniques	Session 1 (2.4.1) – Headers, Footers and Page Numbers	Pages 35-36	
	Session 2 (2.4.2) – Styles and Themes	Page 37-39	
	Session 3 (2.4.3) – Columns, Sections and Links	Pages 39-43	
Topic Five (2.5): Document Review and Collaboration	Session 1 (2.5.1) – Track Changes and Comments	Pages 43-48	
	Session 2 (2.5.2) – Comparing, Merging, Sharing and Permissions	Pages 48-52	
Topic Six (2.6): Bringing it All Together	Session 1 (2.6.1) – Document Design Principles	Pages 52-54	





	Session 2 (2.6.2) – Tips and Advanced Features	Pages 54-58	
ASSIGNMENT TWO	Word Processing	WP skills (primarily practical)	



Module Three: Website Design and Development	Study Sessions	Coursebook Pages	National Curriculum Link
Topic One (3.1): Introduction to Web Development	Session 1 (3.1.1) –Web Development and HTML	Pages 5-8 and 49	Subject content covered in this module: - undertake creative projects that involve selecting, using, and combining multiple applications, preferably across a range of devices, to achieve challenging goals, including collecting and analysing data and meeting the needs of known users - create, re-use, revise and re-purpose digital artefacts for a given audience, with attention to trustworthiness, design and usability - understand a range of ways to use technology safely, respectfully, responsibly and securely, including protecting their online identity and privacy; recognise inappropriate content, contact and conduct and know how to report concerns.
	Session 2 (3.1.2) – Your First Web Page	Pages 10-14	
Topic Two (3.2): HTML and W3C Standards	Session 1 (3.2.1) –W3C Web Standards	Page 15-20	
	Session 2 (3.2.2) – Deeper into HTML	Pages 20-26	
Topic Three (3.3): Styling Basics with CSS	Session 1 (3.3.1) – Introduction to CSS	Pages 26-28	
	Session 2 (3.3.2) – The Box Model	Page 29-30	
Topic Four (3.4): Navigation through Hyperlinks	Session 1 (3.4.1) –Hyperlinks	Pages 30-32	
	Session 2 (3.4.2) – Styling and Attributes	Pages 32-34	
Topic Five (3.5): Engaging with Images and Media	Session 1 (3.5.1) – Image Formats and CSS	Pages 34-37	
	Session 2 (3.5.2) – Embedding Videos	Pages 37-39	
Topic Six (3.6): Styling for Readability and Engagement	Session 1 (3.6.1) – Typography and Images	Pages 39-43	
	Session 2 (3.6.2) – Text Content Layout	Pages 43-48	





ASSIGNMENT THREE	Website Design and Development	Recall and Understanding of definitions and best practices, largely practical design of a site.	
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Module Four: Visual Programming with MIT Scratch	Study Sessions	Coursebook Pages	National Curriculum Link
Topic One (4.1): Getting Started with Scratch	Session 1 (4.1.1) – Welcome to Scratch	Page 1	Subject content covered in this module: - design, use and evaluate computational abstractions that model the state and behaviour of real-world problems and physical systems - use two or more programming languages, at least one of which is textual, to solve a variety of computational problems; make appropriate use of data structures [for example, lists, tables or arrays]; design and develop modular programs that use procedures or functions
	Session 2 (4.1.2) – Getting Started with Scratch	Pages 2-7	
Topic Two (4.2): Sprites, Costumes and Naming Things	Session 1 (4.2.1) – Adding a Sprite	Page 8-11	
	Session 2 (4.2.2) – Seeking Help and Exploring Movement	Pages 11-13	
Topic Three (4.3): Duplicating Sprites	Session 1 (4.3.1) – Duplicating Sprites and Adding Backgrounds	Pages 13- 14	
	Session 2 (4.3.2) – Interacting with the Mouse Cursor	Pages 15-16	
Topic Four (4.4): Interacting with Keypresses and Other Sprites	Session 1 (4.4.1) – Interacting with Keypresses and Other Sprites	Pages 17-19	
	Session 2 (4.4.2) – Key Press Challenges	Page 19	
Topic Five (4.5): Variables	Session 1 (4.5.1) – Creating and Setting Variables	Pages 19-22	
	Session 2 (4.5.2) – Checking Variables and Diverting Program Flow	Pages 22-23	





Topic Six (4.6): Timers and Showing Your Skills	Session 1 (4.6.1) – Timers	Pages 23-24	
	Session 2 (4.6.2) – Exploring and Showing Your Skills	Page 24	
ASSIGNMENT FOUR	Design a Scratch Game	Primarily practical with review.	



Module Five: E-Safety and Using the Web	Study Sessions	Coursebook Pages	National Curriculum Link
Topic One (5.1): Passwords and Searching the Web	Session 1 (5.1.1) – Passwords	Pages 1-3	Subject content covered in this module: understand a range of ways to use technology safely, respectfully, responsibly and securely, including protecting their online identity and privacy; recognise inappropriate content, contact and conduct and know how to report concerns.
	Session 2 (5.1.2) – Searching the Web	Pages 4-7	
Topic Two (5.2): Knowing What to Trust	Session 1 (5.2.1) – Evaluating Websites and Bias	Pages 7-8	
	Session 2 (5.2.2) – Fake News and Online Reviews	Pages 8-10	
Topic Three (5.3): Keeping It Legal	Session 1 (5.3.1) – Internet Laws	Page 10	
	Session 2 (5.3.2) – Using Content Legally	Pages 11-12	
Topic Four (5.4): Saving Your Results	Session 1 (5.4.1) – Bookmarks, Notes and Images	Pages 12-17	
	Session 2 (5.4.2) – Naming, Organisation and Using the Cloud	Pages 18-21	
Topic Five (5.5): Viruses and their Yacky Friends	Session 1 (5.5.1) – Common types of Malware	Pages 22-23	
	Session 2 (5.5.2) – Anti-malware Tools	Pages 23-24	
Topic Six (5.6): Forums, Groups, E-Mail and Social Media	Session 1 (5.6.1) – Forums, Groups, E-mail and Social Media	Pages 24-25	
	Session 2 (5.6.2) – Etiquette and Staying Safe	Pages 26-27	



ASSIGNMENT FIVE	Staying Safe Online	Theory assessment – recall and a long answer question on advice and guidance.	
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Module Six: Getting Started with Arduino Project	Study Sessions	Coursebook Pages	National Curriculum Link
Topic One (6.1): Getting Started and Building Your First Circuit	Session 1 (6.1.1) – Light up an LED	Pages 1-4	Subject content covered in this module:
	Session 2 (6.1.2) – Flash an LED	Pages 5-6	



Topic Two (6.2): Responding to a Button	Session 1 (6.2.1) – Breadboards and Buttons	Pages 7-8	- design, use and evaluate computational abstractions that model the state and behaviour of real-world problems and physical systems - use two or more programming languages, at least one of which is textual, to solve a variety of computational problems; make appropriate use of data structures [for example, lists, tables or arrays]; design and develop modular programs that use procedures or functions - understand the hardware and software components that make up computer systems, and how they communicate with one another and with other systems - undertake creative projects that involve selecting, using, and combining multiple applications, preferably across a range of devices, to achieve challenging goals, including collecting and analysing data and meeting the needs of known users
	Session 2 (6.2.2) – Sensing Buttons	Pages 9-10	
Topic Three (6.3): Latching Buttons Using Variables	Session 1 (6.3.1) – Simple Latches	Page 11	
	Session 2 (6.3.2) – Nesting and more Complex Latches	Page 12	
Topic Four (6.4): Analogue Input and Output	Session 1 (6.4.1) – Analogue Input and Serial Output	Pages 13-14	
	Session 2 (6.4.2) – PWM Output	Pages 15-16	
Topic Five (6.5): Using Analogue to our Advantage	Session 1 (6.5.1) – Subroutines	Pages 17-19	
	Session 2 (6.5.2) – More Efficient Subroutines	Page 20	
Topic Six (6.6): Making a Useful System	Session 1 (6.6.1) – Temperature and Light Sensors	Pages 21-22	
	Session 2 (6.6.2) – Motors and NPN Transistors	Pages 22-24	
ASSIGNMENT SIX	Arduino Traffic Lights	Primarily practical making traffic lights	

